

Question Paper

Exam Date & Time: 29-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIFTH SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026

SUBJECT: CSE 3124/CSE_3124 COMPUTER NETWORKS

(SPL: COMPUTER SCIENCE & ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Consider sending a packet from a source host to a destination host over a fixed route. List the delay (5) components in the end-to-end delay. Which of these delays are constant and which are variable? Justify.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 1B) Using a timing diagram, illustrate and compare the interaction that occur in client-server (3) communication using both TCP and UDP.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 1C) Analyze the devices in the network core and explain how it enables data to travel efficiently from a (2) source to a destination across the Internet.

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) The SSthreshold value for TCP Tahoe TCP station is set to 16. Initial CWND=1 and TCP starts with (4) slow start phase and timer times out during 8th RTT. Represent the problem using proper graph and answer the following:
(a) CWND size during 3^d, 7th, 8th, 11th and 12th RTT.
(b) SSthreshold value at 5th and 10th RTT.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 2B) Apply your understanding of TCP operation to describe how each of the TCP timers (3) (Retransmission, Persistence, Keep-Alive, and Time-Wait) function during connection management and data transfer, using suitable examples or timing scenarios.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 2C) Discuss the importance of the following variables with respect to receiver buffer in achieving TCP (3) flow control between sender A and receiver B LastByteRead and LastByteRcvd. Also, mention how TCP controls overflow through these variables.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3A) Assess the suitability of Link-State (LS) and Distance-Vector (DV) routing algorithms for large, (4) dynamic networks. Explain the advantages and limitations of each and conclude which is more appropriate.

- 3B) Consider the network shown as in Fig 3B. Suppose AS3 and AS2 are running OSPF for their intra-AS routing protocol. Suppose AS1 and AS4 are running RIP for their intra-AS routing protocol. Assume the ASes are configured to share external routing information with each other. (3)

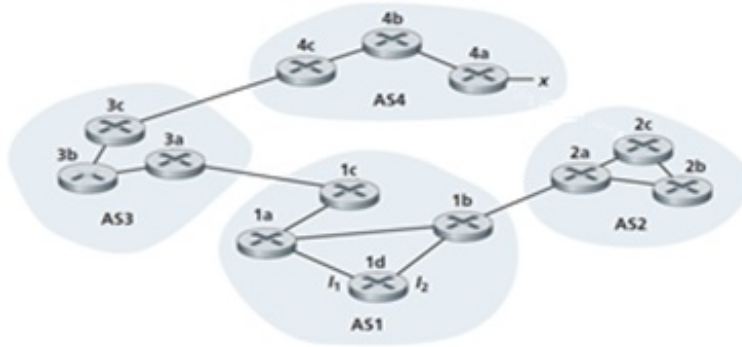


Fig3B

- (a) How does router 3a learn about prefix x?
 (b) Which protocol does router 3a use to distribute information about prefix x within the same AS?
 (c) Router 4c learns about x from which routing protocol?

- 3C) Given a NAT-enabled router with a single public IP and multiple internal hosts, explain how NAT handles multiple simultaneous outbound connections to the same external web server. (3)

- 4A) An organization has been assigned the prefix 212.1.1.0/24 and wants to form subnets for four departments, with hosts as follows: (4)

- (i) 75 hosts
- (ii) 35 hosts
- (iii) 20 hosts
- (iv) 18 hosts

There are 148 hosts in all.

Formulate a possible arrangement of subnet masks to make this possible, and list the subnets in CIDR notation. Propose what the organization might do if department D grows to 32 hosts.

- 4B) A sender sends the data: 11001100 , 10101010 , 11110000, 00001111. Employ a checksum mechanism and provide the transmitted message to the receiver. (3)

- 4C) Show how the Count to Infinity problem may happen with distance vector routing and not with Link State Routing. Use suitable explanations. (3)

- 5A) Given the generator function $G(x) = 1001$ and the message function $M(x) = 11010110$. Compute the codeword. Consider that the codeword is corrupted, such that the receiver receives $R(x) = 101110001111$. Apply mechanisms to detect errors. (5)

- 5B) Demonstrate with a timing diagram how frames are handled when window size is 2^m in case of Go Back N protocol. (3)

- 5C) Outline the concept of vulnerable time. Analyze that the vulnerable time of slotted ALOHA is half of that of pure ALOHA. (2)

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_3124:Computer Networks

#	Course Outcome	Description	Marks
1	1	CO 1	2
2	2	CO 2	5
3	3	CO 3	13
4	4	CO 4	10
5	5	CO 5	10
Total			40

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026

CSE_3124:Computer Networks

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	7	24(60%)
4	Analysis	5	16(40%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	40 (100%)